Non-Invasive Detection of Anemia Using Deep Learning on Conjunctival Images

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***Abstract*—Anemia, characterized by low levels of red blood cells or hemoglobin, affects millions worldwide, significantly affecting public health. Traditional diagnostic methods, while effective, are invasive, costly, and inaccessible in resource-constrained settings. This paper proposes a non-invasive approach for anemia detection using conjunctival images analyzed through deep learning techniques. The proposed methodology involves capturing high-resolution conjunctival images, pre-processing them, and using a customized Convolutional Neural Network (CNN) for feature extraction and classification. The results achieved by the customized CNN fine-tuned with a batch size of 16 give an Accuracy of 96%, Precision of 95%, Recall of 96%, and ROC-AUC score of 0.99. The customized CNN outperformed the other models for this work, such as Random Forest, XGBoost, SVM, ResNet50, and MobileNetV2. This work highlights the potential for non-invasive diagnostic tools to improve accessibility and efficiency in healthcare, particularly for underserved populations. The findings endorse integrating deep learning in healthcare as a transformative approach to address global challenges such as anemia.**

***Index Terms*—Anemia, Conjunctival, Convolutional Neural Network, Random Forest Classifier, XGBoost, Non-invasive**

I. Introduction

Anemia, as defined by the World Health Organization (WHO), is characterized by reduced hemoglobin levels or red blood cell counts below standard thresholds. Anemia remains a persistent public health concern affecting approximately 1.62 billion individuals worldwide - 4.8% of the global population. Symptoms include fatigue, impaired cognitive and physical functioning, weakened immunity, and, in severe cases, life-threatening complications. Vulnerable populations such as children, pregnant women, and individuals in low-income regions experience its effects most acutely. In India, for instance, anemia impacts over half of women and children, underscoring its critical healthcare implications [1].

Conventional anemia diagnostics rely on invasive blood tests that require specialized infrastructure, trained personnel, and reagents—factors that limit their accessibility and affordability in resource-constrained settings. Consequently, there is a pressing need for alternative, cost-effective, and non-invasive diagnostic approaches. One promising avenue is the analysis of conjunctival images: the conjunctiva, a vascular tissue beneath the eyelid, correlates with hemoglobin levels and offers a non-invasive diagnostic window [2].

The COVID-19 pandemic [3] further exposed vulnerabilities in global healthcare systems, often hindering routine diagnostics and aggravating undiagnosed conditions such as anemia. This scenario highlighted the urgent need for diagnostic tools that are accurate, easily deployable, and sustainable in low-resource environments.

Anemia's widespread prevalence and the logistical and financial burdens of traditional blood tests necessitate innovative solutions. In regions like India, where the condition is highly prevalent, non-invasive diagnostics could significantly reduce undiagnosed cases and related complications.

This study addresses the challenges of traditional anemia diagnostics [4] by leveraging advancements in Deep Learning and medical image analysis. By capturing high-resolution conjunctival images and processing them with a Customized Convolutional Neural Network (CNN), the proposed method eliminates the need for invasive sampling, complex laboratory infrastructure, and costly consumables. Complementary feature extraction techniques, combined with integrating models such as Random Forest, enhance the balance between diagnostic accuracy, interpretability, and scalability. Fine-tuning the Customized CNN with a batch size of 16 achieves state-of-the-art performance, delivering 96% accuracy, 95% precision, 96% recall, and a ROC-AUC score of 0.99. Comparative evaluations with other models, including XGBoost, ResNet50, and MobileNetV2, confirm the superiority of the Customized CNN approach. This work marks a significant stride toward accessible, efficient, and scalable anemia diagnostics, offering tangible benefits for underprivileged populations.

This paper is structured into four sections. The first section is RELATED WORK, which reviews relevant literature. The second section is PROPOSED METHODOLOGY, which details Anemia detection methodology. This is followed the section RESULTS AND ANALYSIS which discusses the results obtained by the trained models. The last section CONCLUSION summarizes the paper.

II. Related Work

A. Background and Related Studies

The integration of Machine Learning (ML) and Artificial Intelligence (AI) into healthcare has transformed disease diagnosis and management. Recent advancements in non-invasive

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diagnostic techniques, particularly for anemia, highlight the potential of ML in enhancing accessibility and accuracy.

Ahsan et al. (2022) reviewed ML-based disease diagnosis, emphasizing its role in early detection across various medical fields [5]. Elgohary et al. (2022) explored anemia screening using conjunctival images captured via smartphones, demonstrating the feasibility of non-invasive approaches in resource-limited settings [6]. Çuvadar and Yılmaz (2023) applied deep learning for hemoglobin estimation from conjunctival images, showcasing its potential as a reliable alternative to traditional methods [7]. Kumar et al. (2023) highlighted the importance of diverse datasets in training deep learning models for anemia detection [8]. Singh et al. (2023) proposed a hybrid CNN-SVM model, emphasizing the benefits of ensemble methods for robust diagnostics [9].

TABLE I
COMPARISON OF EXISTING STUDIES ON ANEMIA DETECTION USING ML

Study	Model(s) Used	Accuracy (%)
[6]	KNN, Logistic Regression, Decision Tree, SVM	84.6 (KNN)
[10]	Multiple Regression	83.3
[11]	CNN	85.0

These studies highlight the growing potential of ML in non-invasive disease diagnosis, particularly using image data, and stress the need for continued research in this domain.

B. Outcome of Literature Review

Key findings from the literature include:

- Efficacy of ML Models: Advancements in ML and deep learning have significantly improved diagnostic accuracy in medical applications.
- Non-Invasive Techniques: Using conjunctival images for anemia detection demonstrates the potential of non-invasive diagnostics to replace traditional methods.
- Data-Driven Insights: Comprehensive datasets enhance model performance and generalizability, underscoring the importance of quality data in medical ML research.
- Emerging Trends: The use of smartphones for medical imaging highlights the potential for portable, cost-effective diagnostic tools.

C. Problem Statement

Anemia remains a global health challenge, with invasive and infrastructure-dependent diagnostics limiting accessibility. Developing a reliable, non-invasive, and cost-effective diagnostic method using conjunctival images can address these limitations and improve early detection in resource-limited settings.

III. PROPOSED METHODOLOGY

A. Dataset Preparation and Preprocessing

The dataset used in this study consisted of high-quality conjunctival images captured from patients representing diverse demographic groups. These images were collected using high-resolution cameras (minimum 12MP) under controlled lighting conditions to ensure consistent quality. To avoid glare and reflections, no spotlights or harsh lighting were used during image capture. Medical professionals recorded metadata, including hemoglobin (Hb) levels, age, sex, and anemia classification, alongside the images.

To ensure diversity in the dataset, images were sourced from ten hospitals in Ghana, including Komfo Anokye Teaching Hospital (Kumasi), Bolgatanga Regional Hospital (Bolgatanga), and Sunyani Municipal Hospital (Sunyani). The initial dataset consisted of 710 labeled images, with 426 labeled as anemic and 284 as non-anemic. To address the limited size of the dataset and enhance the model's ability to generalize across varying conditions, the dataset was augmented using various techniques, such as rotation, flipping, scaling, brightness, and contrast adjustments. This augmentation process resulted in an expanded dataset of 4,260 images, ensuring greater variability and robustness for training the Deep learning models.

Fig. 1. Raw Conjunctival Image Captured in a Clinic

Preprocessing Steps:

- **Region of Interest (ROI) Extraction:** A triangle thresholding algorithm was applied to isolate the conjunctiva from the surrounding area, ensuring that only relevant regions were analyzed while removing noise from the background.
- **Image Resizing and Normalization:** Each image was resized to 128×128 pixels to standardize the input dimensions for the machine learning models. Pixel values were normalized to a range of $[0, 1]$, enabling faster model convergence during training.
- **Data Splitting:** The dataset was divided into training, validation, and testing subsets in an 80:10:10 ratio. Specifically, 3,406 images were used for training, 427 for validation, and 427 for testing. This split ensured sufficient data for model training while providing a reliable evaluation of performance on unseen samples.
- **Addressing Class Imbalance:** The Synthetic Minority Oversampling Technique (SMOTE) was applied to the

training data to generate synthetic samples for the minority class - non-anemic (based on the dataset), thereby balancing the dataset without duplication [12].

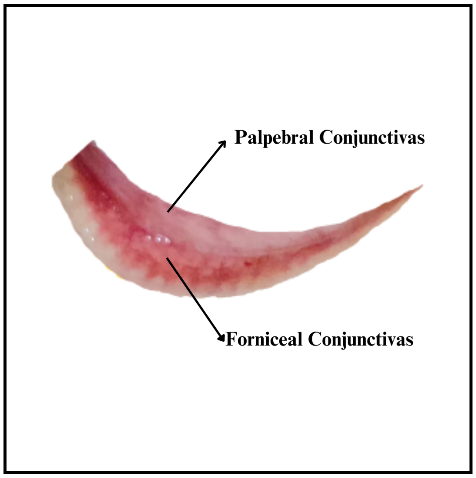

Fig. 2. Extracted Region of Interest (ROI) for Analysis

These preprocessing steps ensured that the dataset was well-prepared for training robust machine learning models, minimizing noise, addressing class imbalance, and standardizing the input for optimal model performance.

B. Customized CNN Architecture

The Customized Convolutional Neural Network developed for this study was explicitly designed to analyze conjunctival images for anemia detection. Unlike generic pretrained models, this CNN was designed to capture hierarchical spatial features that are critical for anemia detection, such as variations in conjunctival pallor and vascular patterns. By incorporating domain-specific insights into its architecture, the Customized CNN effectively balances computational efficiency and predictive accuracy.

The Customized CNN comprises four convolutional layers, each followed by max-pooling layers that progressively reduce the spatial dimensions of the feature maps. This hierarchical structure allows the model to preserve essential features while eliminating irrelevant details. The convolutional layers use ReLU (Rectified Linear Unit) activation functions, which introduce non-linearity and enable the model to learn complex feature representations.

A fully connected dense layer aggregates the high-level features extracted by the convolutional layers. This layer, combined with a dropout rate of 30%, helps mitigate overfitting by preventing the model from relying too heavily on specific neurons during training. The final dense layer uses a sigmoid activation function, outputting a probability score for binary classification (anemia vs. non-anemia).

The architecture is summarized in Table II, detailing the output dimensions, kernel sizes, and parameter counts for each layer.

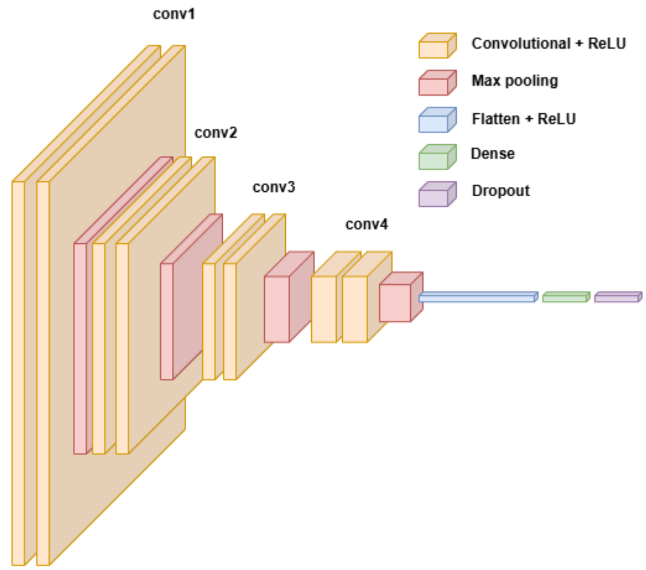

Fig. 3. Customized CNN Architecture

TABLE II
CUSTOMIZED CNN ARCHITECTURE SPECIFICATIONS

Layer Type	Output Shape	Kernel Size	No. of Parameters
Input	$128 \times 128 \times 3$	-	0
Conv2D + ReLU	$126 \times 126 \times 32$	3×3	896
MaxPooling2D	$63 \times 63 \times 32$	2×2	0
Conv2D + ReLU	$61 \times 61 \times 64$	3×3	18,496
MaxPooling2D	$30 \times 30 \times 64$	2×2	0
Conv2D + ReLU	$28 \times 28 \times 128$	3×3	73,856
MaxPooling2D	$14 \times 14 \times 128$	2×2	0
Conv2D + ReLU	$12 \times 12 \times 256$	3×3	295,168
MaxPooling2D	$6 \times 6 \times 256$	2×2	0
Flatten	$9,216$	-	0
Dense + ReLU	256	-	2,359,552
Dropout (30%)	256	-	0
Dense + Sigmoid	1	-	257

1) Key Design Features: The Customized CNN exhibits several key design features that make it particularly well-suited for this task:

- Task-Specific Architecture: Tailored for conjunctival images, capturing both low-level textures and high-level patterns.
- Lightweight and Efficient: Fewer parameters than deep pretrained models, suitable for deployment on resource-constrained devices.
- Robust Regularization: Dropout prevents overfitting, ensuring better generalization.

2) Deployment Suitability: Due to its lightweight design and efficient architecture, the Customized CNN is suitable for deployment in real-time diagnostic systems, including on portable devices such as smartphones or embedded medical systems. This makes it an ideal candidate for use in resource-constrained or remote settings where traditional diagnostic methods may not be feasible.

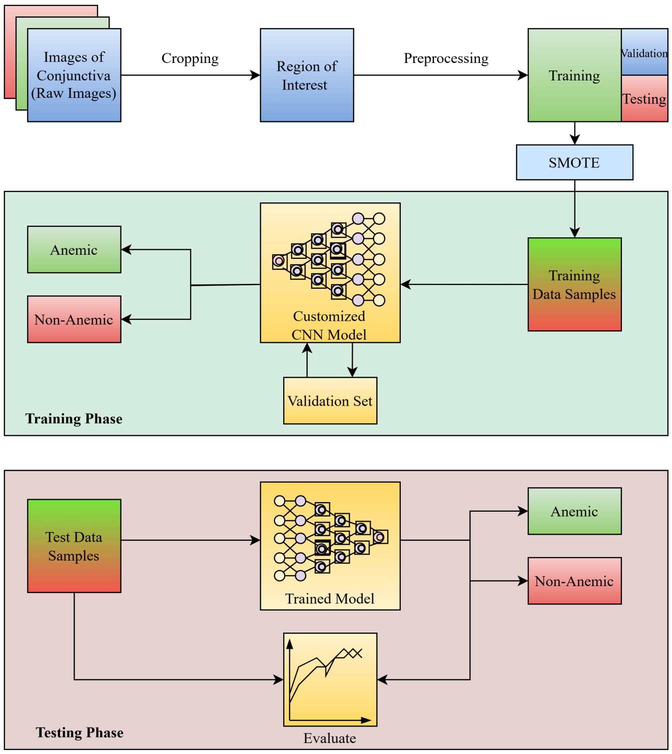

Fig. 4. Flow Diagram of Proposed Methodology

C. Model Training and Fine-Tuning

The training of the Customized CNN was carried out in two carefully planned stages, each designed to maximize model performance and generalization capability through the strategic use of different batch sizes and optimization techniques.

1) **Initial Training (Batch Size 8):** The model was initially trained with a smaller batch size of 8. This approach allowed for more precise gradient updates, which is particularly useful when working with datasets that exhibit high variability. Training with a smaller batch size ensured that the model could effectively learn fundamental patterns within the data, even in the presence of noise. However, the smaller batch size also introduced higher fluctuations in validation loss due to less stable gradient estimates. These fluctuations underscored the need for subsequent fine-tuning to achieve robust convergence and improve performance stability.
2) **Fine-Tuning (Batch Size 16):** In the fine-tuning phase, the batch size was increased to 16 to provide smoother and more stable gradient updates. This stability was critical during the later stages of training, as the model required subtle parameter adjustments to refine its decision boundaries. The larger batch size also reduced noise in gradient updates, enabling the model to generalize better to unseen data. This phase optimized the model's performance, ensuring that it captured nuanced patterns in the input data while maintaining resilience to variability.

D. Special Features of the Customized CNN

The Customized CNN incorporates several unique features that make it particularly well-suited for the task of non-invasive anemia detection:

- **Domain-Specific Feature Extraction:** The architecture is specifically designed to capture critical features such as conjunctival pallor and vascular patterns, which are essential for anemia detection.
- **Scalability:** The model's efficient architecture and training protocol make it adaptable to larger datasets and diverse imaging conditions, ensuring scalability across various applications.
- **Explainability:** Hierarchical feature extraction layers provide interpretable insights into the model's predictions, facilitating clinical adoption by enhancing trust and understanding among healthcare professionals.
- **Lightweight Design:** The streamlined architecture reduces computational overhead, making the model suitable for deployment in resource-constrained settings or on portable devices.

E. Considering Pretrained Models

To explore alternative approaches to anemia detection, pretrained models such as ResNet50 and MobileNetV2 were fine-tuned for this work. These models were selected due to their prominence in image classification tasks and their lightweight yet robust architectures.

- **ResNet50:** ResNet50, a deep residual network with 50 layers, is designed to address vanishing gradient issues by incorporating skip connections that allow gradients to flow through deeper layers. This enables the network to capture hierarchical features effectively. For this work, ResNet50 was initialized with pretrained ImageNet weights, and a Customized classification head was added. The classification head included a global average pooling layer, a dropout layer (30%), and a dense output layer with sigmoid activation. Fine-tuning was performed by unfreezing the last 30 layers to adapt the model to the specific features of conjunctival images.
- **MobileNetV2:** MobileNetV2 is a lightweight convolutional neural network optimized for efficiency in mobile and embedded systems. It uses depthwise separable convolutions and linear bottlenecks to reduce computational overhead while maintaining accuracy. Similar to ResNet50, MobileNetV2 was initialized with pretrained ImageNet weights and augmented with a Customized classification head. During initial training, the base layers were frozen to leverage pretrained features, and fine-tuning was performed by unfreezing the final layers to adapt the network to the dataset.

These models provided benchmarks for comparison and insights into the challenges of using general-purpose architectures for domain-specific medical image analysis. The modifications and fine-tuning processes aimed to align these models with the task-specific requirements of anemia detection, highlighting their strengths and limitations.

F. Considering Other Models

In addition to deep learning approaches, traditional machine learning models—Random Forest, XGBoost, and Support Vector Machine (SVM)—were employed. These models utilized features extracted from the penultimate layer of the Customized CNN, transforming raw image data into high-dimensional feature vectors.

- **Random Forest:** Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy. For this work, the model was configured with 100 trees, each trained on bootstrapped subsets of the feature data. The majority voting mechanism enabled robust classification, especially for diverse patterns in the dataset.
- **XGBoost:** XGBoost is a gradient-boosted decision tree algorithm that uses additive learning and regularization techniques to prevent overfitting. Its feature importance mechanism provided insights into the most influential features for anemia classification. Hyperparameters such as the number of estimators, learning rate, and maximum depth were optimized using grid search to ensure the best fit for the extracted feature space.
- **Support Vector Machine (SVM):** SVM is a linear model that aims to find the hyperplane that best separates classes in a high-dimensional space. For this work, the radial basis function (RBF) kernel was used to handle nonlinear feature interactions. The model was configured with balanced class weights to address the inherent dataset imbalance.

These models served as baselines to compare with the end-to-end deep learning approaches. By utilizing extracted CNN features, they demonstrated the potential of combining feature extraction and traditional classifiers in medical image analysis.

G. Model Setup in the Work

The pretrained and traditional models were integrated into the work as follows:

- **Pretrained Models:** The training process began by freezing the convolutional base during the initial epochs to utilize general features effectively, followed by selectively unfreezing layers for fine-tuning. This two-step approach facilitated task-specific adaptation while minimizing the risk of overfitting.
- **Traditional Models:** Feature vectors from the second-to-last layer of the Customized CNN were flattened and used as input. Hyperparameter tuning was performed using grid search for both Random Forest and XGBoost, while SVM relied on kernel-based classification without extensive hyperparameter optimization.
- **Evaluation Pipeline:** All models were trained and validated on the same dataset split for consistency. SMOTE-resampled data was applied to ensure balanced learning across all models.

This setup provided a comprehensive evaluation framework, enabling insights into the suitability of each model type for non-invasive anemia detection.

IV. Results and Analysis

A. Performance of Customized CNN

The Customized CNN demonstrated exceptional performance after fine-tuning with a batch size of 16. Following training, the model was evaluated on the test set consisting of 427 images. This model effectively captured the unique features of conjunctival images, achieving state-of-the-art metrics as follows:

- **Accuracy:** 96%
- **Precision:** 95%
- **Recall:** 96%
- **F1 Score:** 95%
- **ROC-AUC:** 0.99

The confusion matrix in Fig. 5 highlights the model's ability to classify anemic and non-anemic samples accurately, with minimal errors. Additionally, the training and validation curves in Fig. 6 exhibit consistent improvement in accuracy and loss convergence, showcasing the model's robust generalization capabilities.

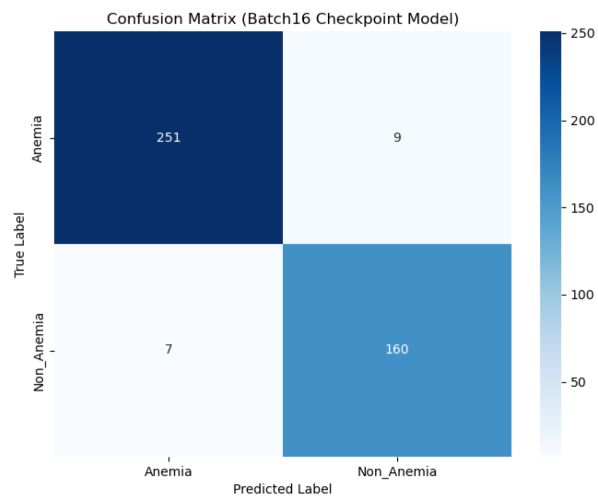

Fig. 5. Confusion Matrix (Customized CNN, After fine-tuning)

B. Comparison with Pretrained Models

The training of the Customized CNN was conducted in two carefully planned stages, focusing on utilizing different batch sizes and optimization techniques to maximize model performance and generalization capability. Each phase played a critical role in refining the model's learning process.

- **ResNet50:** The model achieved an accuracy of 77%, with its strong residual connections helping to learn hierarchical features. However, its large parameter count increased computational requirements, making it less practical for deployment in resource-constrained settings.
- **MobileNetV2:** This lightweight model achieved an accuracy of 73% but struggled to generalize effectively due to its reduced feature extraction capacity. Its simplicity,

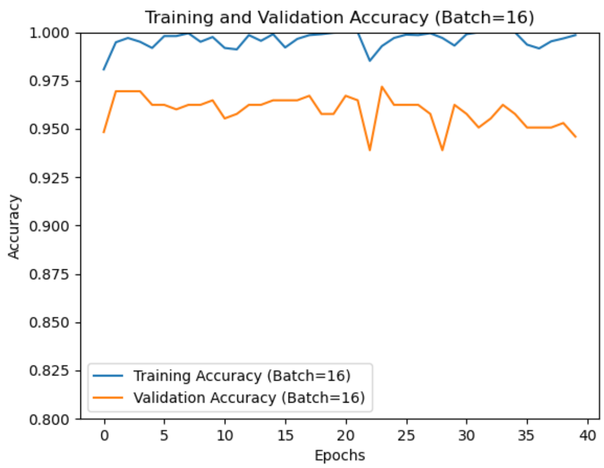

Fig. 6. Training and Validation Accuracy (On fine-tuning)

while computationally efficient, limited its ability to capture subtle patterns unique to conjunctival images.

C. Comparison with Traditional Models

Traditional machine learning models, including Random Forest, XGBoost, and SVM, were evaluated using features extracted from the penultimate layer of the Customized CNN. These models provided insights into how well handcrafted feature representations could perform in comparison to end-to-end learning approaches.

- **Random Forest:** Achieved an accuracy of 92%. The ensemble-based decision tree approach performed robustly but showed slightly lower recall, indicating a tendency to misclassify some anemic samples.
- **XGBoost:** Performed slightly better than Random Forest, achieving 93% accuracy. Its gradient-boosted framework effectively captured feature interactions but was outperformed by the Customized CNN in overall metrics.
- **Support Vector Machine (SVM):** Achieved an accuracy of 79%. The SVM struggled with the high-dimensional feature space and dataset imbalance, making it less effective for this task.

D. Summary of Model Performance

The performance of all evaluated models is summarized in Table III. The Customized CNN significantly outperformed pretrained and traditional models across all key metrics. Its ability to learn task-specific features directly from raw data gave it a distinct advantage, resulting in higher accuracy, recall, and overall robustness.

The results clearly demonstrate that the Customized CNN is the most effective model for non-invasive anemia detection, surpassing both pretrained and traditional models. Its lightweight architecture and superior metrics make it ideal for deployment in real-world medical settings, particularly in resource-limited environments.

TABLE III
MODEL PERFORMANCE COMPARISON

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Random Forest	92	92	87	90
XGBoost	93	93	89	91
SVM	79	74	72	73
ResNet50	77	78	85	82
MobileNetV2	73	72	52	61
Customized CNN	**96**	**95**	**96**	**95**

E. Error Analysis

While the Customized CNN achieved high performance, a small number of misclassifications were observed, primarily in borderline cases where conjunctival features were less distinguishable. These errors are attributed to:

- Variability in image quality, lighting, and patient conditions.
- Overlap in visual features between mildly anemic and non-anemic cases.

Future work can address these issues by incorporating larger datasets, advanced data augmentation techniques, and explainability tools such as Grad-CAM to better understand model predictions.

V. Conclusion

This study presents a novel, non-invasive approach for anemia detection using conjunctival images analyzed through deep learning and machine learning techniques. By utilizing domain-specific insights and a custom-designed Convolutional Neural Network, the proposed methodology achieves state-of-the-art performance, significantly outperforming pretrained models such as ResNet50 and MobileNetV2, as well as traditional classifiers like Random Forest, XGBoost, and Support Vector Machine.

The Customized CNN, tailored specifically for conjunctival image analysis, demonstrated an accuracy of 96%, precision of 95%, recall of 96%, and an ROC-AUC score of 0.99 after fine-tuning. These results highlight the network's ability to extract and utilize hierarchical features critical for detecting anemia with minimal misclassifications. The model's lightweight architecture and task-specific optimization make it suitable for deployment in real-time diagnostic systems, particularly in resource-constrained environments where traditional blood-based testing is impractical.

References

[1] A. Sinha, M. Adhikary, J. P. Phukan, S. Kedia, and T. Sinha, "A study on anemia and its risk factors among pregnant women attending antenatal clinic of a rural medical college of west bengal," *Journal of Family Medicine and Primary Care*, vol. 10, no. 3, pp. 1327–1331, Mar 2021, epub 2021 Apr 8.

[2] T. N. Sheth, N. K. Choudhry, M. Bowes, and A. S. Detsky, "The relation of conjunctival pallor to the presence of anemia," *Journal of General Internal Medicine*, vol. 12, no. 2, pp. 102–106, Feb 1997.

[3] A. Kakde, D. Sharma, and N. Arora, "Optimal classification of covid-19: a transfer learning approach," *Int. J. Comput. Appl*, vol. 176, no. 20, pp. 25–31, 2020.

[4] A. Kakde, N. Arora, and D. Sharma, "Novel approach towards optimal classification using multilayer perceptron," *International Journal of Research in Engineering, IT and Social Sciences*, vol. 810, pp. 29–38, 2018.
[5] M. Ahsan *et al.*, "Machine-learning-based disease diagnosis: A comprehensive review," *Healthcare*, vol. 10, no. 3, 2022.
[6] S. Elgohary *et al.*, "A machine learning method to screen anemia from conjunctiva images taken by smartphone," in *2022 JAC-ECC*, 2022, pp. 125–128.
[7] B. Çuvadar and H. Yılmaz, "Non-invasive hemoglobin estimation from conjunctival images using deep learning," *Medical Engineering Physics*, vol. 120, 2023.
[8] R. Kumar *et al.*, "Deep learning for anemia detection using conjunctival images," *IEEE Access*, vol. 11, pp. 30 215–30 224, 2023.
[9] P. Singh *et al.*, "Hybrid machine learning model for anemia detection from conjunctival images," *J. Med. Imaging Health Informatics*, vol. 13, no. 6, pp. 1218–1228, 2023.
[10] N. Sehar and K. Nirmala, "Analysis of conjunctiva for screening of anemia," in *2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI)*. IEEE, 2024, pp. 1–6.
[11] M. Ramkumar, V. Aarthi, and G. Yuvaraj, "Deep convolutional neural network model for concurrent prediction of anemia and blood glucose level from conjunctiva images," in *2024 International Conference on Automation and Computation (AUTOCOM)*. IEEE, 2024, pp. 434–438.
[12] D. Dablain, B. Krawczyk, and N. V. Chawla, "Deepsmote: Fusing deep learning and smote for imbalanced data," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 9, pp. 6390–6404, 2023.